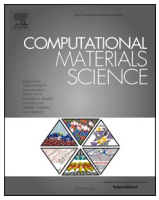
Computational Materials Science 239 (2024) 112984



Contents lists available at ScienceDirect

Computational Materials Science

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Full Length Article



Chemical library generation of polymer acceptors for organic solar cells with higher electron affinity

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ARTICLE INFO

Keywords:
Machine learning
Organic solar cells
Electron affinity
Descriptors

ABSTRACT

In this study, an intricate machine learning assisted framework is introduced for the designing of polymer acceptors. Machine learning (ML) models are trained to predict the electron affinity of polymers. Ten thousand new polymers are generated using the Breaking Retrosynthetically Interesting Chemical Substructures (BRICS) method and their electron affinity values are predicted using a pre-trained ML model. The selection of polymers is based on electron affinity, retaining those with higher electron affinity. The synthetic accessibility of the selected polymers is evaluated. Additionally, the structural similarity among the selected polymers is examined; results are indicating higher similarity between selected compounds. By efficiently identifying and optimizing new polymers, the approaches developed significantly enhance the likelihood of discovering superior materials for advanced applications.

1. Introduction

Solution-processed organic solar cells emerge as a highly promising technology for clean and renewable energy [1,2]. Their lightweight, flexibility, and large area fabrication make them a favorable option for solar power generation. Additionally, the potential for low-cost materials and fabrication processes enhances their appeal [3]. Over the recent years, there has been extensive research on polymers owing to their adaptable chemical and physical properties. The diverse functional groups present in polymers allow for the regulation of morphologies at both bulk and interface levels [4,5]. Their high carrier mobilities make polymers suitable for functioning as materials for electron and hole transfer [6]. The chances of structures and functional modifications also provide polymers with a range of optical absorption properties and variable electron mobility, making them valuable as photo-active layers in organic solar cells [7].

The synthesis and characterization of new polymers require substantial funds and manpower, and occasional serendipity. The brute-force approach may not always yield polymers with the desired properties, necessitating a potential adjustment of the design threshold or facing overall design failure [8]. Consequently, the discovery or design of new polymers often originates from well-equipped and renowned laboratories possessing ample domain knowledge, talented researchers, financial resources, time, and manpower [9]. In an effort to expedite polymer design and discovery, a longstanding history exists in leveraging artificial intelligence (AI) to streamline the process [10].

Electron affinity (Eea) is a foundational concept in chemistry [11]. It represents the energy change that transpires when an atom acquires an electron, resulting in the formation of a negative ion [12]. Essentially, electron affinity quantifies the attraction between an atom and an additional electron. A higher electron affinity corresponds to a greater amount of energy released when an electron is added to the atom.

In semiconductors, electron affinity plays a crucial role in determining whether the material exhibits n-type or p-type conductivity. N-type materials, serving as the majority charge carriers, possess an excess of electrons, whereas p-type materials feature an excess of “holes”

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https://doi.org/10.1016/j.commatsci.2024.112984
Received 11 December 2023; Received in revised form 26 March 2024; Accepted 26 March 2024
Available online 29 March 2024
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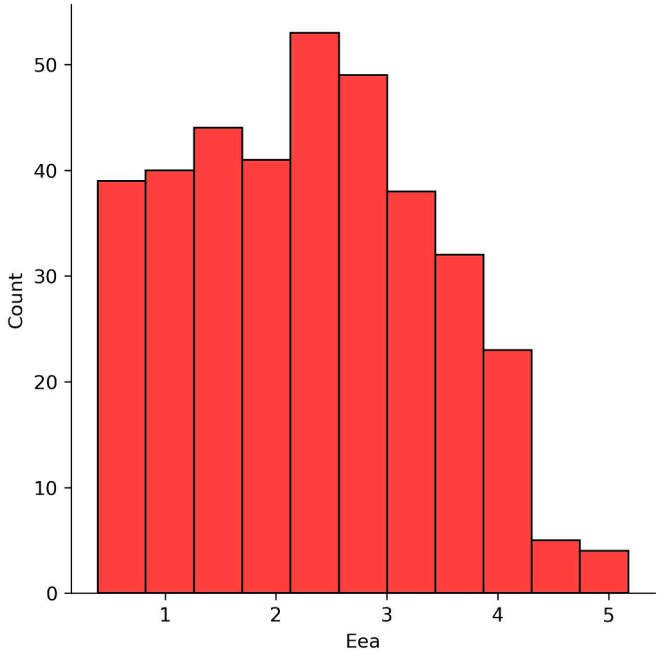


Fig. 1. Distribution of Eea (eV) in data set.

(electron vacancies) [13]. A thorough understanding of electron affinity is vital for the design of new materials with specific electronic properties, particularly for applications in electronic devices. Scientists leverage this understanding to create novel materials tailored for various uses, especially in the development of electronic gadgets. The ability to modify the conductivity, energy band structure, and other electronic properties by adjusting electron affinity empowers researchers to craft more efficient and specialized electronic materials [14,15].

Organic solar cells constitute a photovoltaic technology that utilizes organic semiconductors to convert sunlight into electricity [7,16]. In the performance of organic solar cells, acceptor materials play a pivotal role. Efficient charge separation in organic solar cells also relies on the electron affinity of the acceptor material. To maximize the efficiency of these solar cells, it is crucial to adjust the electron affinities of both the donor and acceptor materials [17,18]. This adjustment is achieved by introducing electron-donating groups to the donor material and electron-withdrawing groups to the acceptor material. The acceptor material, with a higher electron affinity than the donor material, efficiently accepts electrons from the donor material. Understanding the role of electron affinity in organic solar cells is fundamental for designing new materials that offer improved performance and enhanced efficiency in converting sunlight into electricity [19].

Density functional theory (DFT) is the most commonly used tool for the investigation of electronic properties of materials. The problem with the DFT is its high computational cost when dealing with the large system [20]. Another drawback of DFT is its failure to calculate the exact exchange–correlation energy and to counter this problem several approximation methods have been developed. Machine learning (ML) is a subgroup of artificial intelligence (AI) which emphasis on modifying the codes and models having ability to learn from data [21]. Also, the computational cost of the ML is much cheaper than DFT methods. DFT methods are time consuming but ML methods predict rapidly which makes them valuable for specific work. ML models can also attain high perfection with suitable algorithms and data [22].

Speed is the key advantage of ML over DFT. For complex system DFT calculations are computationally exorbitant and time taking. Trained ML models predicts much faster than other regularly used simulation methods, allowing for high-throughput screening of materials [23,24]. In addition, ML can present non-linear relationships in data, without the required of detailed quantum mechanical calculations, ML enables advancement in models for material properties [25,26]. Another edge of ML over DFT is that ML models can be employed on less powerful hardware when once trained. DFT is not a better method for predicting the specific properties of new materials but ML have the ability to provide somehow exact predictions for certain properties of new materials [27,28].

This study presents a data-driven framework for the creation of polymer acceptors, employing machine-learning (ML) models trained to forecast the electron affinity of polymers. The influence of various descriptors on the outcomes of the top-performing ML model is clarified using SHapley Additive exPlanations (SHAP). Subsequently, 10,000 polymers are generated and their electron affinity values are projected using the most effective ML model. The selection of polymers is determined by their electron affinity, with a preference for those exhibiting higher electron affinity. The synthetic accessibility of the selected polymers is then assessed.

2. Computational methods

2.1. Machine learning analysis

Data is wheel of machine learning. Data can be collected from literatures or taken from available database. In present study, electron affinity data of polymers is taken from published paper [29]. Data is consisting of more than 368 data points; electron affinities of polymers were calculated using density functional theory (DFT) calculations. Fig. 1 depicts the distribution of Eea (eV) in a data set. It displays the frequency of various Eea values in the data collection. The Eea values are represented on the x-axis, and the frequency or count of those values is represented on the y-axis. The figure looks to be positively biased, with lower Eea values occurring more frequently and higher Eea values occurring less frequently. For Eea levels between 2 and 3, the greatest frequency is around 40–50 counts. For Eea, the lowest frequency is about 0–10 counts. The conversion of chemical structures into machine-readable form is also important task. In present study, molecular descriptors are used. RDkit is used to calculate molecular descriptors [30]. A library of about 200 descriptors was generated. Calculating a lot of

Table 1
List of important descriptor.

No	Name	Type	Descriptor Class	Description
1	VSA_EState 10	Topological	E-State	Sum of E-State indices for atoms with valence state 10
2	fr_C_S	Structural	Fragment Count	Number of C-S bonds in a molecule
3	SMR_VSA10	Topological	SMR	Sum of topological surface areas for atoms with atomic charge index 10
4	SlogP_VSA12	Topological	SLogP	Sum of log P contributions of atoms with atomic charge index 12
5	Kappa1	Topological	Kappa	First-order Kappa shape index
6	FractionCSP3	Structural	Fractional	Fraction of carbon atoms with sp^3 hybridization
7	SMR_VSA6	Topological	SMR	Sum of topological surface areas for atoms with atomic charge index 6.

descriptors for molecules in a short period of time is simple. Every descriptor was not relevant for the model training. We have reduced their numbers by taking multiple actions. We didn't select descriptors with zero values. Univariate regression analysis was used to find the best descriptors. Selection of the appropriate machine-learning model is a critical step in building effective predictive models. Different algorithms offer various trade-offs in terms of accuracy, computational efficiency, and generalization capabilities. Various python-based packages are used for analysis and visualization such as Seaborn [31], Pandas [32], Numpy [33], Matplotlib [34] and Scikit-learn [35]. Data set is divided into training and test set with 70:30 ratio, respectively.

2.2. Monomer design and similarity analysis

Theoretical designing of polymers is done using RDkit. A set of cheminformatics tools is called RDkit. Breaking Retrosynthetically Interesting Chemical Substructures (BRICS) tool [36] was utilized to obtained desired number of new polymers. The obtained chemical space of polymers were visualized and analyzed. Chemical space is a multi-dimensional space in which molecules can be represented using their chemical properties. To delve into the chemical similarity of the devised polymers, a cluster plot and heatmap were employed. In this analysis, chemical fingerprints were utilized to assess the structural similarities among the designed polymers.

3. Results and discussion

3.1. Molecular descriptors

The behavior of organic molecules in various applications is dictated by their chemical composition [37–39]. Molecular descriptors are basically theoretically derived or experimentally measured properties of a molecule. More particularly, characteristics of a molecule like topological, physical or chemical are quantitatively described by molecular descriptors [40–42]. Molecular descriptors are mathematical representations of molecular properties that can be calculated using a variety of procedures. The numerical values of molecular descriptors are quick and simple to compute. However, quantum chemical descriptors necessitate computational power and time. Molecular descriptors can be used to do similarity analysis (search) in molecular libraries [43–45]. As a result, depending on the descriptor values, these may be useful for screening molecules with comparable chemical and physical properties. They summarize our knowledge and comprehension of molecular structure and exertion from several perspectives. Molecular descriptors represent a molecule's physical and chemical qualities [46–48].

Approximately 200 descriptors are discovered and reduced using various methods. Descriptors with constant or negligible (zero) values are initially excluded. Second, the descriptor pairings with the highest self-correlation are removed. Table 1 contains information about the selected descriptors. These descriptors (features) are utilized to train machine-learning models.

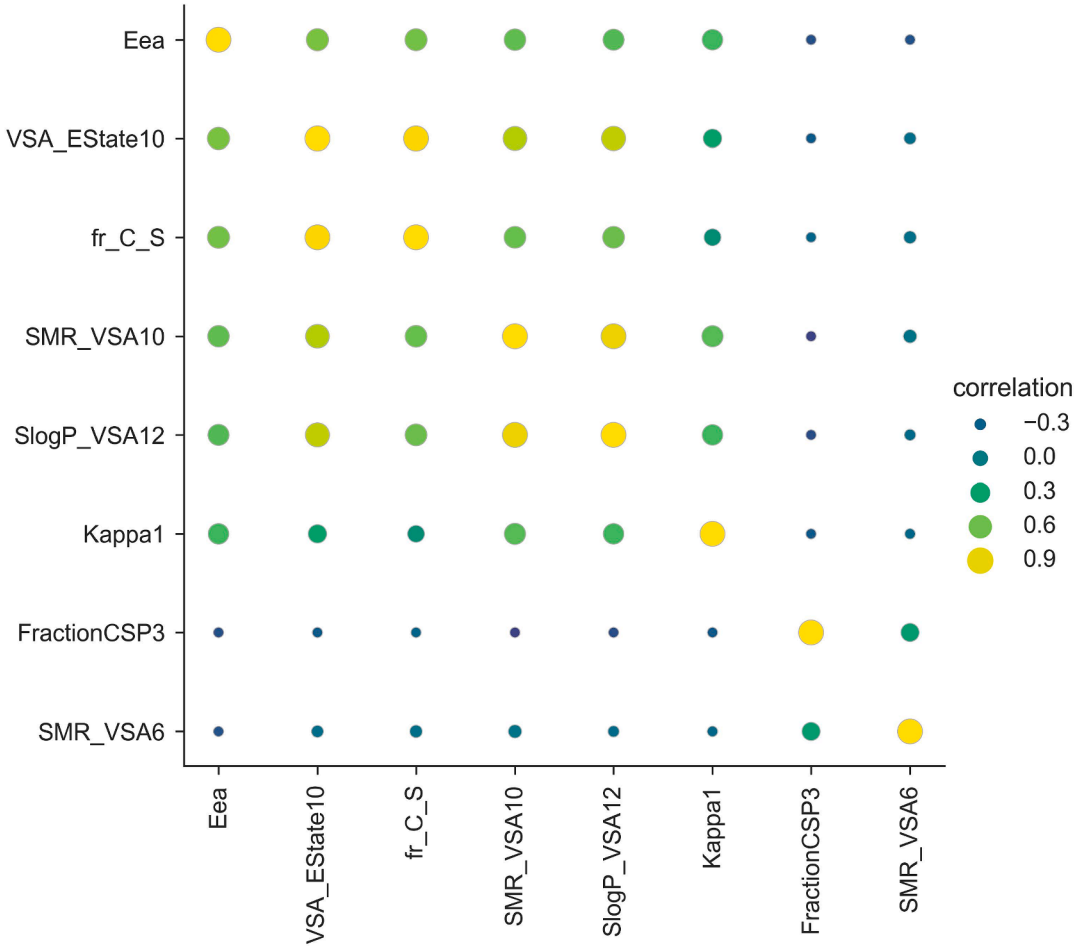


Fig. 2. Correlation between Eea and top descriptors.

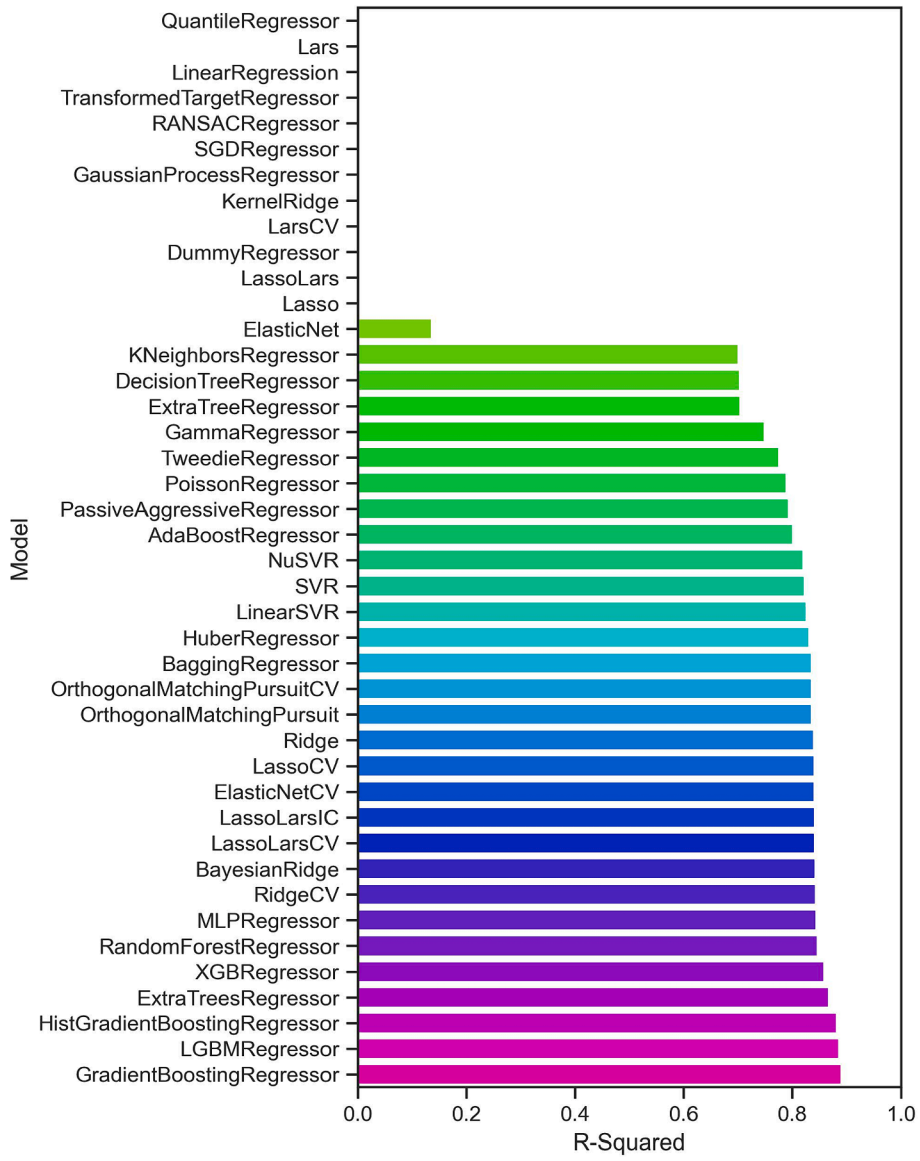


Fig. 3. Comparative assessment of the performance of diverse machine learning models on the test set.

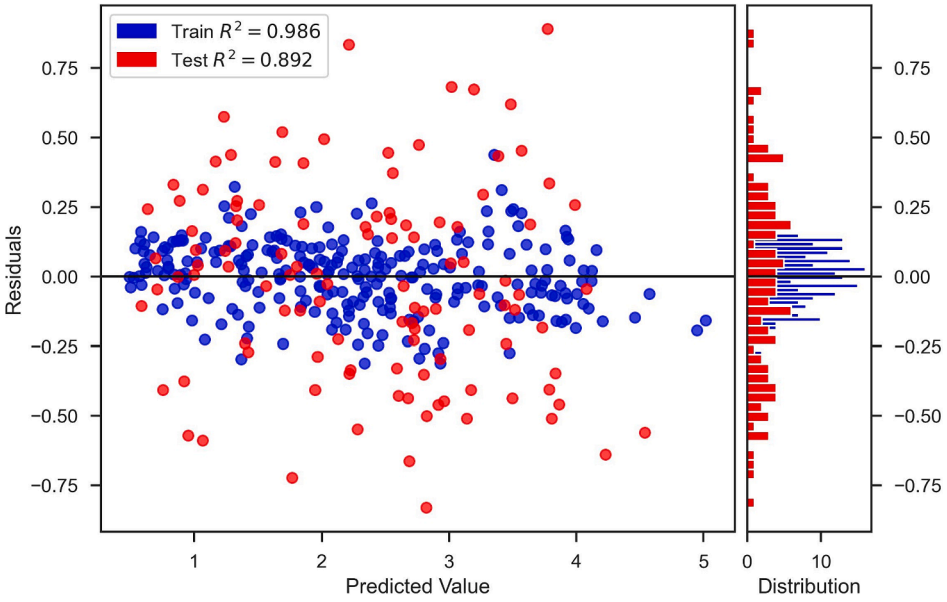


Fig. 4. The plot of residuals for the regression model with gradient booting.

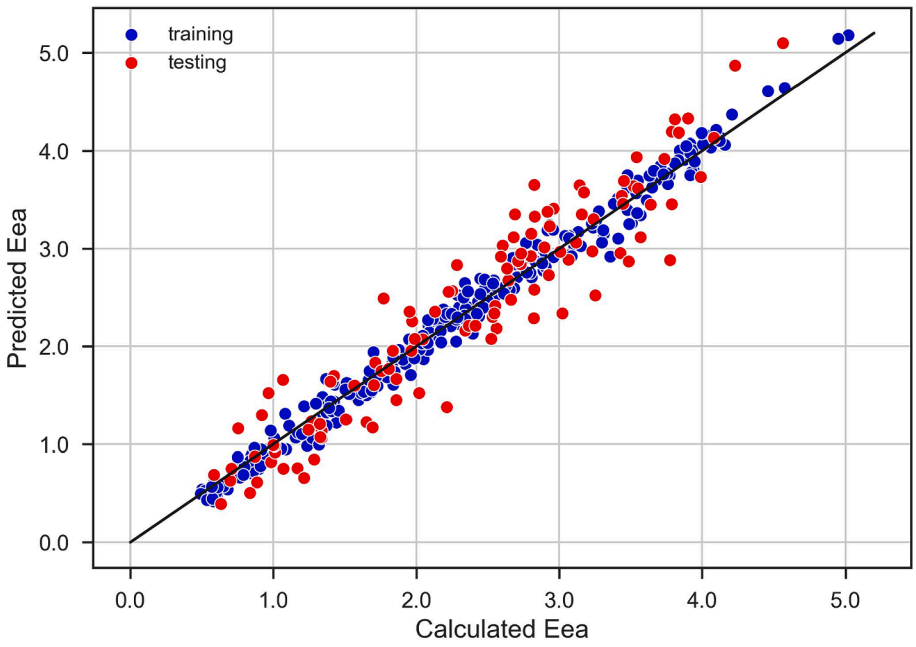


Fig. 5. The scatter plot demonstrated by gradient booting regression between the predicted and calculated Eea.

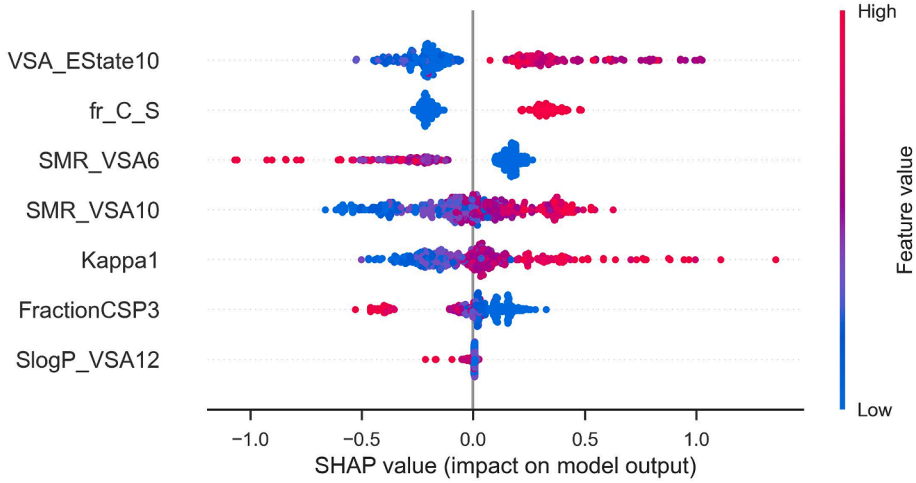


Fig. 6. The effects of certain descriptors, both positive and negative, on the Hist gradient booting regression model's output.

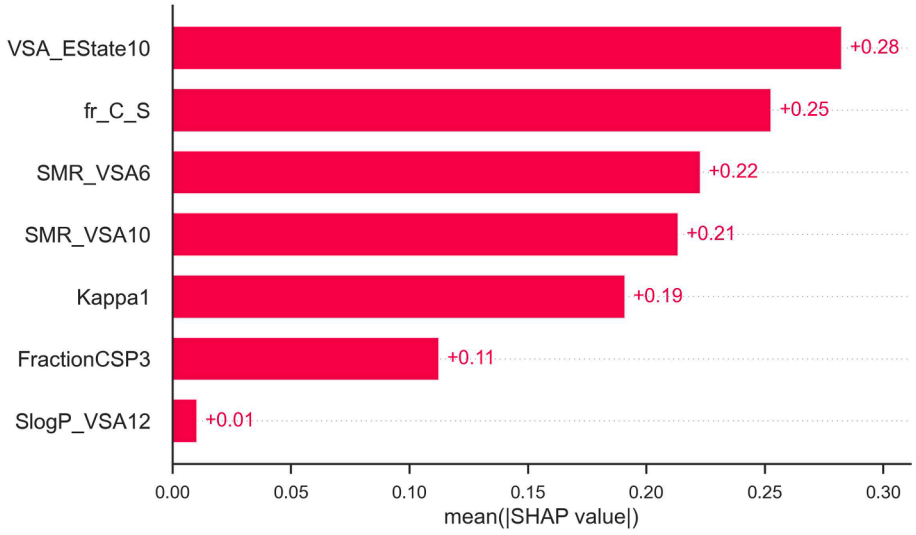


Fig. 7. The average influence of selected descriptors on the results of the gradient booting regression model.

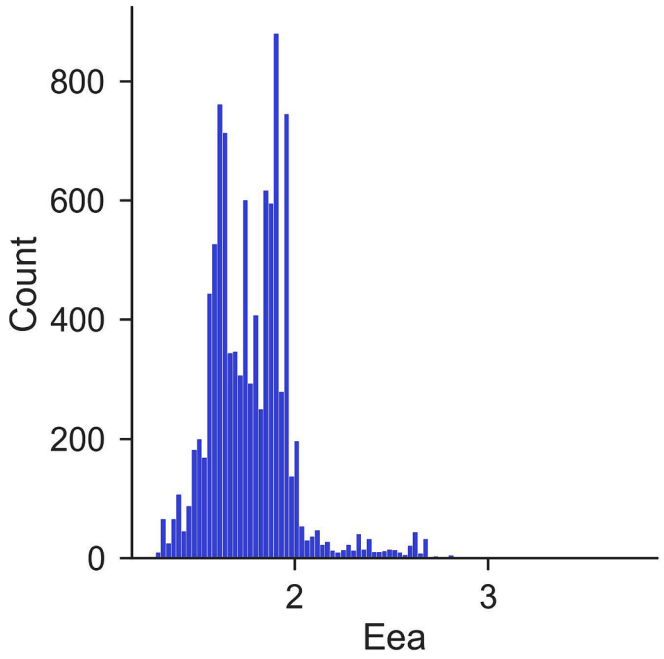


Fig. 8. The distribution plot of estimated Eea (eV) for the generated polymers.

The correlation between Eea and top descriptors is given in Fig. 2. The correlation between descriptors is less than 0.9. Eea has positive correlation with 5 descriptors and negative correlation with 2 descriptors. Highest correlation is 0.6 and lowest correlation is −0.3. This indicates these five descriptors greatly depend upon the Eea of organic molecules and others have no effect with the Eea of organic molecules.

3.2. Regression analysis

The type of model utilized has a significant impact on the output of machine learning models. A python-based code named Lazy Predicted is used to find the best machine-learning model. Lazy Predict is a Python library designed to simplify and accelerate the process of building and evaluating machine learning models. Lazy Predict is designed to integrate with scikit-learn, a comprehensive Python library for machine learning, offering tools for various predictive data analysis tasks [35]. Lazy Predict automates the model selection and evaluation process, providing a convenient and timesaving solution. Lazy Predict contains more than forty models. About 25 models have shown R-squared (R^2) values higher 0.8 (shown in Fig. 3). Gradient boosting regressor has appeared as best model. This selection is done on the basis of R^2 value. Here, we also like to make clear that many new models such as deep learning architectures (Transformers, GPT, BERT), state-of-the-art NLP models (GPT-3, T5), or the latest ensemble methods and models for unstructured data (e.g., EfficientNet, Vision Transformer) are not included in Lazy Predict.

The model's performance is assessed using the relationship between the calculated and predicted Eea values. Analyzing and visualizing the correlation between the calculated and predicted Eea create a scatter plot. Typically, a residual plot is a dispersed graph with the desired value on the horizontal (y-axis) and the residuals on the vertical (x-axis) plane. The disparity between the actual and predicted values is indicated by the residuals' values. The predicted and actual values diverge more from zero the more the data points' values deviate from this value. Generally speaking, the residual values are not very high. There was a prominent peak near zero in the scatter plot. Good R^2 values are those that are close to 1. The residual plot for gradient boosting regressor is given in Fig. 4. Fig. 4 reveals that the R2 value for the test dataset closely aligns with that of the trained dataset, affirming the accuracy of the chosen model without bias. This suggests its suitability as a general model for broader applications.

Scatter plot between calculated and predicted values is given in Fig. 5. Points are closed to standard line. If the values are close to the line of best fit, we can expect the model to perform well and the predicted values to be close to the true values.

3.3. Shapley additive explanations (SHAP)

SHAP analysis provides a means to elucidate the precision of machine learning models by employing a game-theoretic framework, specifically through additive explanations. Utilizing Shapley or SHAP values allows for an analysis of the contributions of individual features in explaining model outputs. The Shapley value, representing the average contribution of each coalition member to the overall coalition value, serves as a key metric in assessing feature importance. To gauge the significance of a feature in a model, one must comprehend how altering that feature influences the model's output and the distribution of values associated with that feature. These values offer insights into whether a particular characteristic has a positive or negative impact on the model's outcomes.

SHAP was utilized to analyze the prediction behavior of the models by calculating the contribution of each feature. Fig. 6 (standardized features data) depicts the interpretation of the GBRT model using SHAP. Fig. 6 illustrates the effect of various features/descriptors on positive and negative SHAP values. Positive values indicate that the specific feature contributed more. Fig. 6 show that the higher the SHAP value for VSA_EState 10, fr_C_S, SMR_VSA10, SlogP, VSA12, Kappa1, FractionCSP3, and SMR_VSA6, the greater the feature effect for the prediction of desired polymers.

Choosing the appropriate descriptors that give the best indication of the model's accuracy is crucial. To make sure of this, we ranked the descriptors according to their decisive role in the model's correctness using the machine learning approach known as extra tree regression (as shown in Fig. 7). Out of the 200 descriptors, only 7 descriptors VSA_EState10, fr_C_S, SMR_VSA10 and SlogP_VSA12 have individual positive values. VSA_EState10 and fr_C_S had the greatest mean SHAP values among them, suggesting that the model's prediction might be significantly reduced if these descriptors were eliminated.

3.4. Designing new polymers

There are numerous synthetic processes for organic substances, whether they are small molecules or polymers [49–51]. Every decision made during the process, whether it adds heteroatoms or substituents, has a substantial impact on the material's structure, optoelectronic properties, and performance. The new materials can be designed using a variety of diverse methodologies [52,53]. We demonstrated a simple and fast method for producing new polymers. The efficient building components are first screened from several databases before being linked together to create innovative polymer materials [54–56]. The development of innovative organic polymers for organic solar cells is a new topic of study. They can be made in a variety of styles. We employed an unusual technique to design novel polymers. New polymers are created by linking the recognized building units [57,58]. Synthetic materials are taken into account during this process.

Designing new polymers involves choosing polymers, finding out polymerization methods and modifying molecular structure for particular properties [59–61]. Different polymers have different properties.

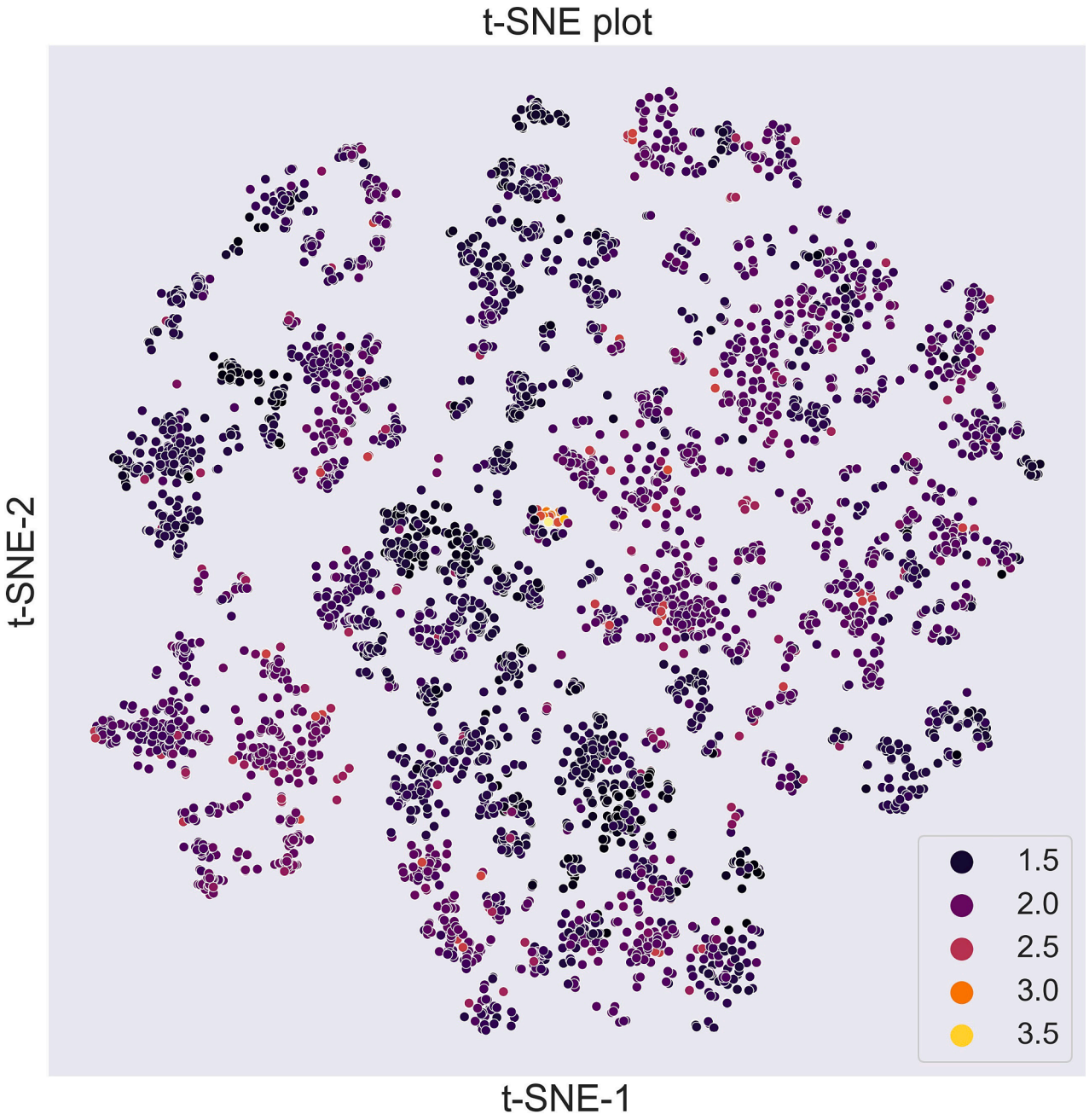


Fig. 9. T-sne plot.

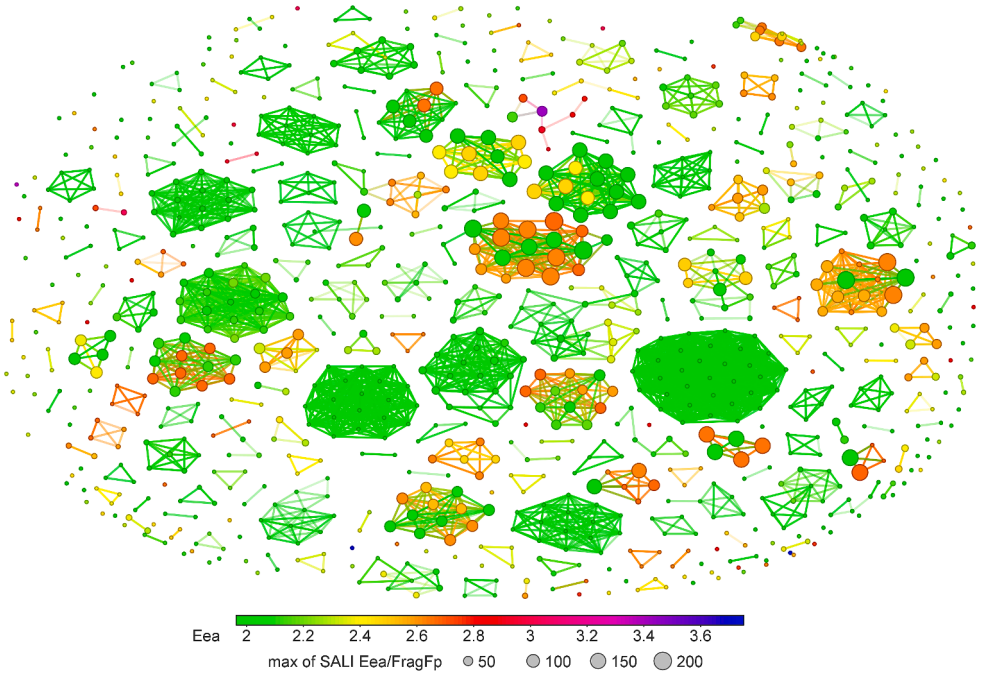


Fig. 10. SALI plot for top 1000 polymers with highest electron affinity.

Fig. 8 shows that the majority of developed polymers have an electron affinity of 1–2, indicating that they can easily receive electrons and can be employed as good acceptors in organic solar cells.

A t-Distributed Stochastic Neighbor Embedding (t-SNE) plot is a scatter plot that is used to depict high-dimensional data in a lower-dimensional space. The goals of this graphic are to find patterns and clusters in the data as well as to understand the underlying structure of the data. It is extensively used in data science and machine learning to analyze and visualize complex datasets. When dealing with a much larger or a much more complex data set t-SNE find a way to project data into a low dimensional space. Clusters of alike data points are many a time indicative of fundamental patterns or structures in the data set. t-SNE plot is displayed in Fig. 9. Lower and higher Eea values are distributed everywhere in generated space.

The Structure Activity Landscape Index (SALI) plot is a type of graph used to depict the relationship between two variables. It is used to identify patterns or trends as well as to understand the relationship between variables. The SALI plot technique, which provides a SALI score, can be used to find structure–activity cliffs, or pairs of molecules with the greatest resemblance but the biggest activity change.

The structure actively landscape index (SALI) is a concept employed in computational chemistry to evaluate the connection between activities and the chemical structures of compounds. SALI is designed to grab the topography of the structure–activity within a dataset. SALI helps to understand the changes in compounds activities by varying their molecular structure.

SALI analysis is computationally expensive, therefore, one thousand polymers with higher Eea values are selected and results are presented in Fig. 10. Polymers are divided into large number of groups. Large number of small rings is indicating less change in Eea values on structural change in majority of cases. Majority of generated polymers are similar and showing lower Eea values.

We have chosen 30 polymers. Figs. 11–13 depict their structures. In nature, the chosen chemicals are structurally scattered. Our method is useful for discovering novel chemicals.

We have performed density functional theory (DFT) calculations on the 3 polymers. B3LYP/6–31 g(d, p) level of theory is used. First three polymers have shown the values of 2.82, 2.79 and 2.61 eV, respectively. There is notable difference between machine learning predicted and DFT calculated values. However, DFT is not suitable for large-scale screening. Therefore, relatively lower accuracy is acceptable for preliminary screening. Machine learning can be used in the initial step of screening, then DFT can be applied to a limited number of compounds. Therefore, machine learning is a viable option due to its speed and ability to screen a large number of compounds.

3.5. Synthetic accessibility

Following the generation of the chemical library, a pivotal step involves assessing or predicting the relative synthetic accessibility scores of the generated chemical database. Synthetic accessibility is critical for commercialization of organic solar cells [62]. Its computation involves summing fragment scores and applying a complexity penalty. The fragment score is derived from statistics on the frequency of Extended Connectivity Fingerprints of diameter 4 (ECFP4) [63] fragments, generated using Pipeline Pilot [64], on nearly one million molecules sourced from the PubChem database [65]. ECFP is a method that numerically represents a chemical structure by traversing it, enumerating atoms, and hashing their representation. The complexity penalty is introduced to assess if a molecule contains too many intricate structures that may pose challenges during synthesis. It considers factors such as the number of aromatic rings, stereocenters, macrocycles, and the overall size of the molecule. The higher the synthetic accessibility score, the more challenging it becomes to synthesize that particular compound, and conversely, lower scores indicate ease of synthesis. Synthetic accessibility score values range from 01 (easy/facile synthesis) to 10 (difficult synthesis), with a threshold at 6 to distinguish between easy and challenging synthesis. Threshold is suggested by authors of paper

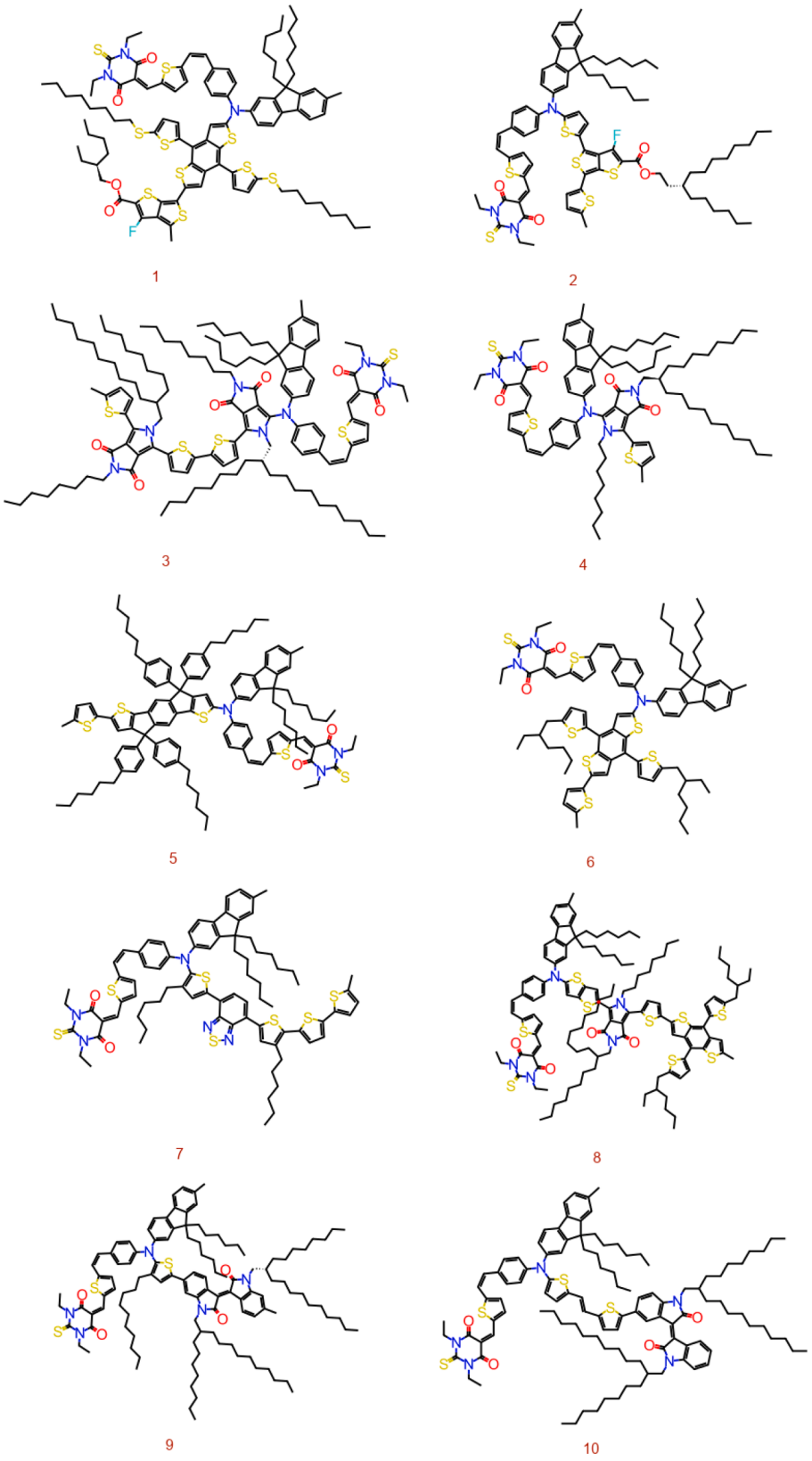


Fig. 11. Top 1–10 designed polymers with higher predicted Eea values.

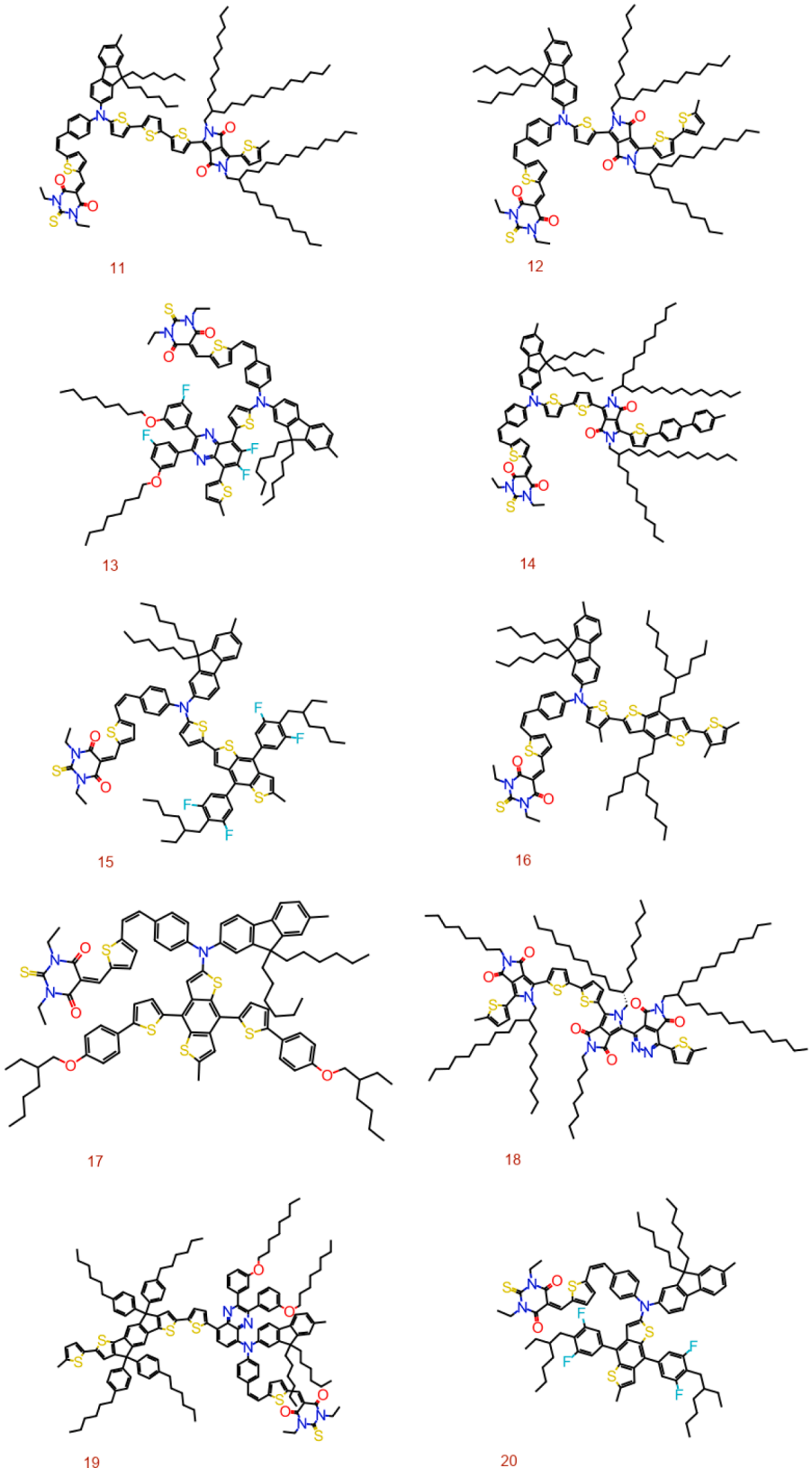


Fig. 12. Top 11–20 designed polymers with higher predicted Eea values.

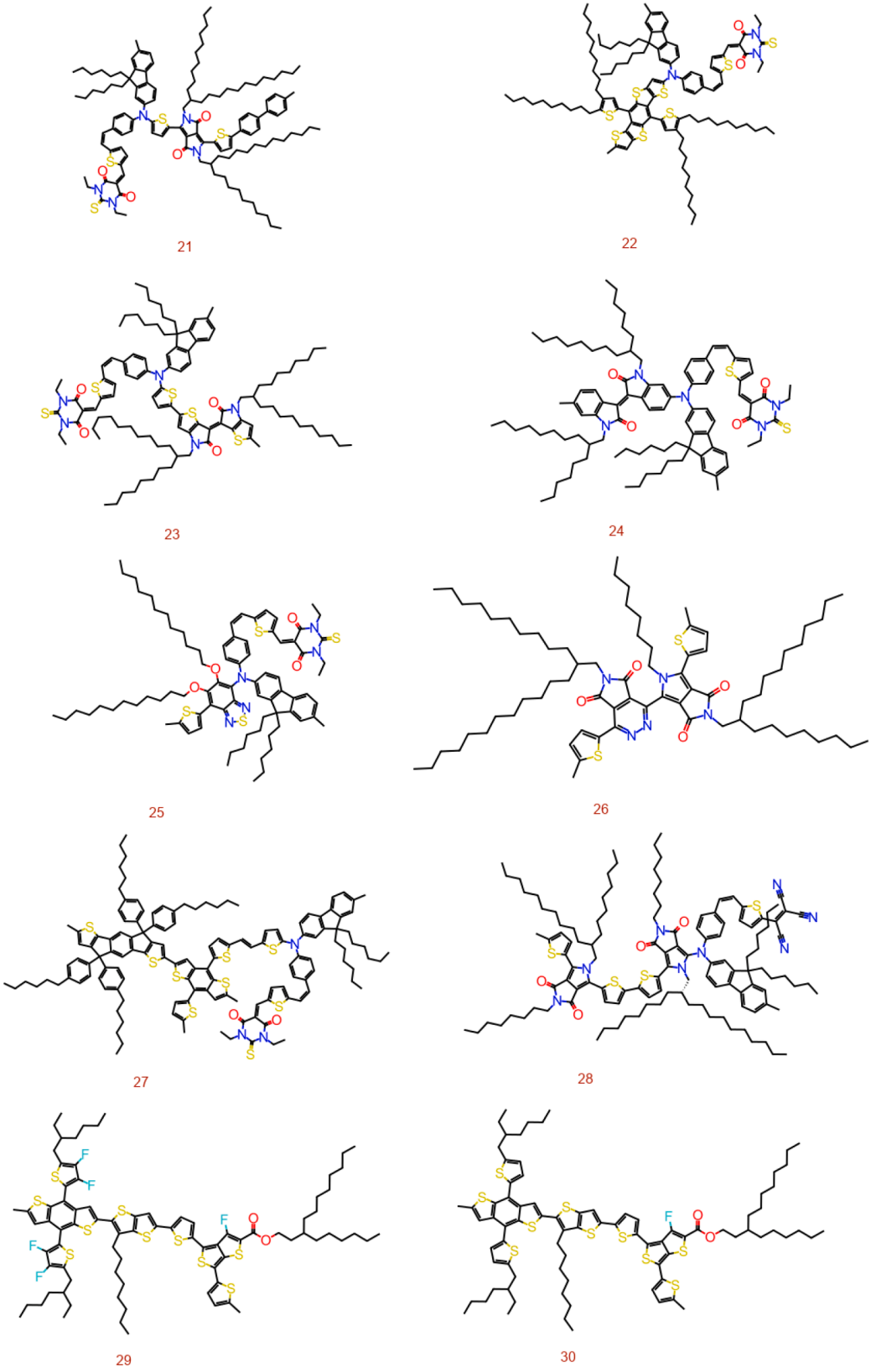


Fig. 13. Top 21–30 designed polymers with higher predicted Eea values.

Table 2
The predicted electron affinity (Eea) values along with the synthetic accessibility (SA) score of the top thirty designed polymers.

Name	Eea (eV)	SA score
1	3.75	5.53
2	3.69	4.54
3	3.43	6.51
4	3.37	4.56
5	3.04	5.04
6	3.04	5.04
7	3.02	4.2
8	2.98	6.94
9	2.95	5.79
10	2.94	5.7
11	2.93	5.96
12	2.93	5.53
13	2.89	4.56
14	2.89	6.13
15	2.86	5.21
16	2.83	5.24
17	2.82	5.21
18	2.82	5.81
19	2.82	6.78
20	2.81	5.03
21	2.8	5.93
22	2.79	5.11
23	2.76	5.56
24	2.75	4.91
25	2.73	4.03
26	2.73	3.96
27	2.72	6.21
28	2.71	6.35
29	2.69	5.47
30	2.69	5.15

published in 2009 [66]. As detailed in Table 2, the majority of the monomers exhibit synthetic accessibility score values below 06, signifying that these compounds are easy to synthesize and demonstrate higher practical applicability. This observation underscores the accessibility and feasibility of synthesizing the designed monomers, enhancing their potential for practical use.

3.6. Similarity analysis

Fig. 14 illustrates the clustering of polymers based on their structural similarities, achieved through the utilization of similarity-based clustering algorithm. These algorithms primarily rely on similarity or distance metrics to group together data points that share similarities, while isolating those that are dissimilar or distant into distinct clusters. In the case of this exploratory data analysis, chemical structures can be arranged using similarity-based clustering algorithms [67]. Polymers are divided in to three clusters. One cluster contains majority of polymers. Polymers are not significantly different from each other. All the cluster connect each other near to distance of 0.4.

Fig. 15 depicts a heatmap illustrating the relative similarity among the designed polymers, utilizing a color-coded scheme for data visualization. The color pattern of the heatmap effectively demonstrates the distribution of polymers using a similarity approach. Lowest correlation value is 0.70, it is indicating that selected polymers are highly similar. Majority of polymers have shown correlation values near to 0.85.

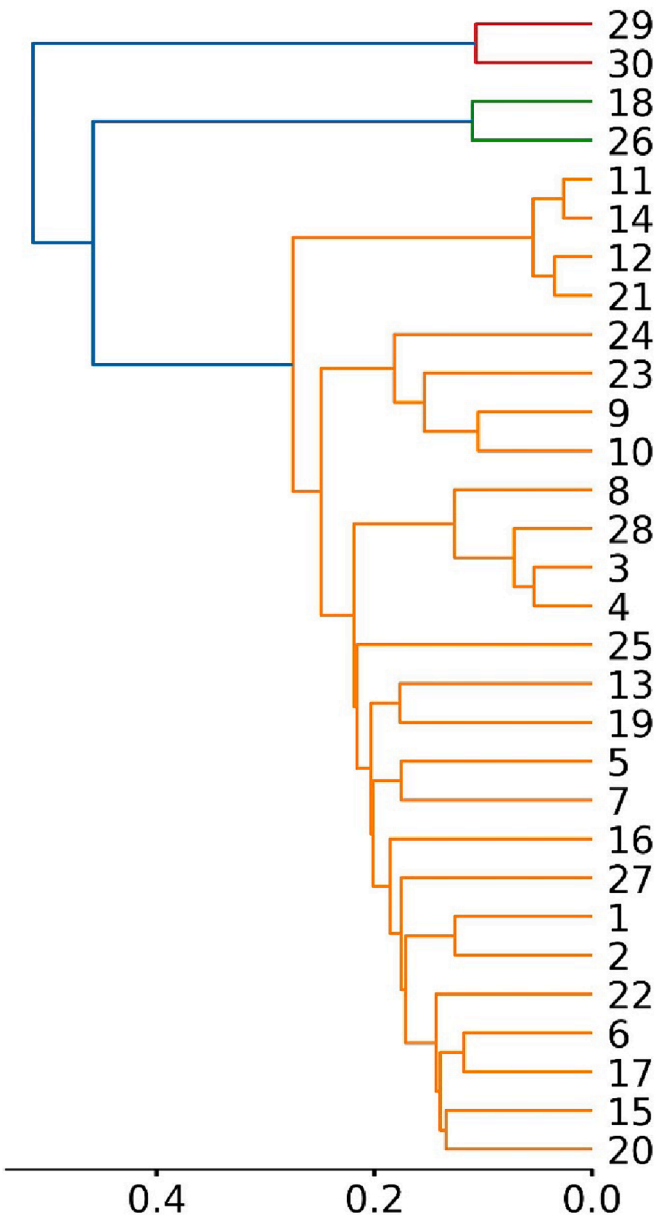


Fig. 14. Clustering of polymers based on similarity analysis.

4. Conclusions

Efficient material design demands innovative and cost-effective approaches to accelerate the discovery of superior materials within constrained timelines. In this study, we have used machine-learning models to predict the electron-affinity of polymers. A substantial database of polymers was generated, and these machine learning models, having been previously trained, were employed to predict the crystallization tendency of the newly generated polymers. Various data visualization tools were employed to unveil the structural diversity inherent in these compounds. To gauge the synthetic accessibility of the chosen compounds, we conducted calculations, revealing that all identified compounds can be easily synthesized. A chemical similarity-clustering analysis indicated a noteworthy degree of structural similarity among the selected compounds. The proposed framework stands out as a valuable tool for strategically selecting the most effective compounds in the material design process, offering a streamlined and informed approach to material discovery.

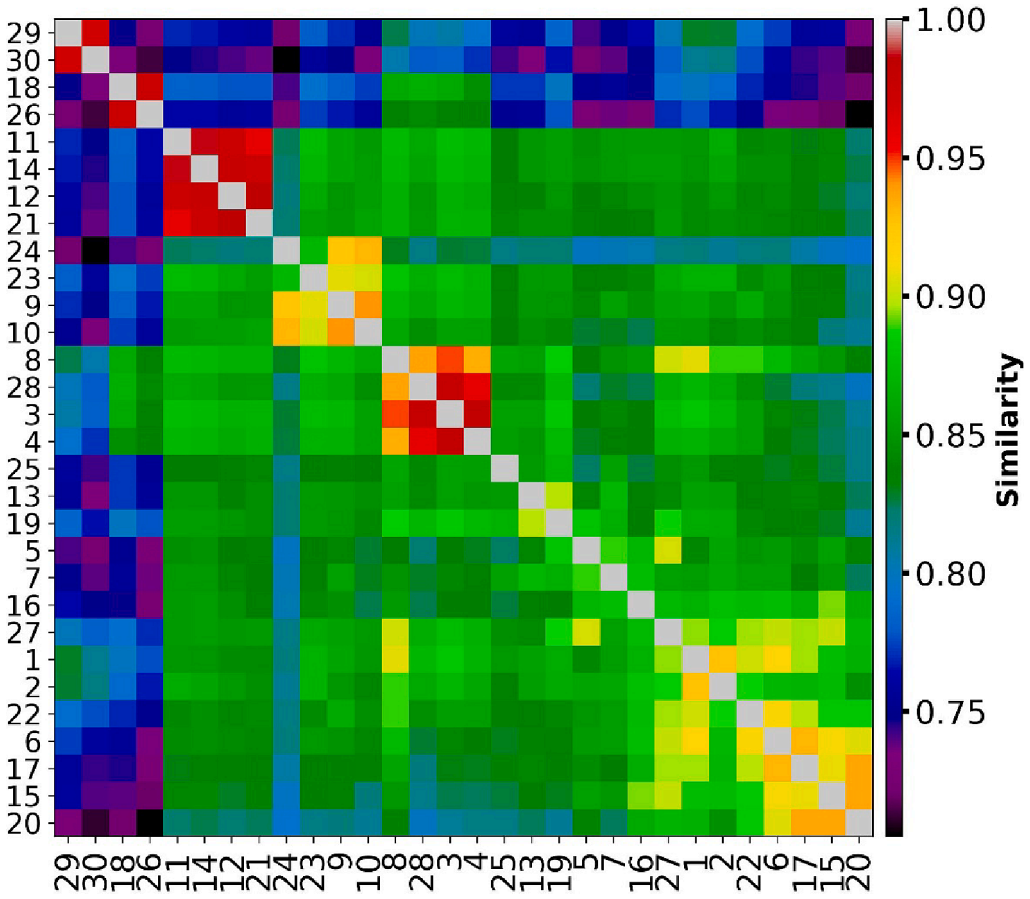


Fig. 15. Heatmap of similarity between designed polymers.

CRediT authorship contribution statement

Fatimah Mohammed A. Alzahrani: Writing – original draft, Investigation. **Sumaira Naeem:** Investigation, Data curation. **Numan Khan:** Software, Formal analysis. **Bilal Siddique:** Writing – original draft, Data curation. **Muhammad Faizan Nazar:** Writing – review & editing, Supervision, Conceptualization. **Tagir Kadyrov:** Visualization, Project administration, Funding acquisition. **Z.A. Alrowaili:** Writing – review & editing. **M.S. Al-Buriahi:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

The authors express their gratitude to Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2024R42), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia.

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